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CSE437: DATA SCIENCE: CODING WITH REAL-WORLD DATA

Quiz #1

Student ID: **Date**: July 15, 2026

Name: **Marks**: 20

Time: 60 min

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1. You are processing a real estate dataset to train a machine learning model. The dataset contains a feature for `House_Price`. While most house ranges between $150,000 and $500,000, a few luxury mansions in the dataset are valued at over $20,000,000.
   If you want to scale this feature, which scaling should you use and why? [2]
   a. Min-max scaling
   b. Standard scaling
   c. Robust scaling

2. An insurance company is reviewing its current policy rates. When originally setting the rates, they believed that the average claim amount will be maximum $180,000. They are concerned that the true mean is actually higher than this, because they could potentially lose a lot of money. They randomly select 40 claims, and calculate a sample mean of $195,000. Assume that the standard deviation of claims is $50,000. Setting significance level, $\alpha=0.05$, test to see if the insurance company should be concerned or not. [6]

Significance level, α	10%	5%	1%
$Z_{1-\alpha}$	1.28	1.645	2.33
$Z_{1-\alpha/2}$	1.645	1.96	2.33

3. Consider the following transaction database collected from a retail store, where each transaction represents items purchased together:

TID	Items Bought
100	*f, a, c, d, g, i, m, p*
200	*a, b, c, f, l, m, o*
300	*b, f, h, j, o*
400	*b, c, k, s, p*
500	*a, f, c, e, l, p, m, n*

Using the FP-growth algorithm, generate all frequent itemsets from the above transaction set with minimum support count of 3. Your answer must show the complete simulation. [12]